COLE DYCK

coled.dev@gmail.com • linkedin.com/in/cole-dyck/ • github.com/Zemya

PROFESSIONAL SUMMARY

Computer Science graduate (University of the People, June 2026) focused on data engineering with hands-on experience building backend applications, data pipelines, relational databases, and cloud-based deployments in Linux and Azure. Strong foundation in Python, SQL, Linux, relational databases, and software design. Seeking entry-level data engineering or backend software engineering role.

EDUCATION

Bachelor of Science, Computer Science | University of the People, USA *June 2026*

- Cumulative GPA: 3.67 — Honors List (Sep 2024), Dean's List (Spring & Fall 2025, Spring 2026)

PROJECTS

AI Job Matching Automation | n8n, Docker, Gemini API, REST APIs | *2026*

- Built a scheduled n8n workflow, self-hosted in Docker, that queries Adzuna (five role searches) and RemoteOK in parallel branches and merges the results
- Used the Gemini API to analyze my resume, score each posting against it, and draft tailored cover letters; filtered to matches scoring 85 or higher
- Cleaned and deduplicated results by job URL, logging them to Google Sheets with Pending status tracking and email detection
- Ran under a 4 GB memory limit, with a migration to a home Ubuntu server planned

Algorithmic Trading Bot (investBot-Py) | Python, Alpaca API, SQLite, Linux | *2026*

- Built a Python bot that polls one-minute market data every 60 seconds and trades EMA5/EMA10 crossover signals with a volume filter, using Alpaca's paper trading API
- Rebuilds a daily watchlist from Yahoo Finance's most-active list (price and ticker filters, with a fallback list) and sizes each position at 2% of account cash
- Applies take-profit (1.5%), stop-loss (1%), and momentum exit rules, and flattens all positions before market close
- Logs trades to SQLite and events to file, with psutil memory monitoring and a main loop whose exception handling keeps it running
- Runs unattended as a background process (nohup) on a headless Ubuntu server; configuration via python-dotenv

Information Retrieval System | Python, SQLite, NLTK | *2025–2026*

- Developed a web crawler and search engine using BeautifulSoup for HTML parsing and NLTK's Porter Stemmer for text normalization.
- Implemented a text preprocessing pipeline including tokenization, stop-word removal, and stemming.
- Designed an inverted index in SQLite and implemented TF-IDF weighting to rank documents based on query relevance.

NYC Taxi Data Pipeline | Python, PostgreSQL, PyArrow, SQLAlchemy | *2025*

- Built a Python pipeline that streams a 3.48M-row NYC TLC Parquet file into PostgreSQL in 100,000-row batches, keeping memory use bounded
- Queried the loaded data with SQL on a Linux server and visualized results in Python
- Documented the project with a README and requirements.txt, and wrote a database connection test script

Stirling-PDF Cloud Deployment | Docker, Azure | *2024*

- Deployed containerized web application on Microsoft Azure using Docker with environment configuration and persistent storage

- Configured cloud networking, monitoring, and resource tracking using Azure Monitor
- Gained hands-on experience with container lifecycle management and cloud deployment workflows

TECHNICAL SKILLS

- **Languages:** Python, Java, C#, SQL, R, PHP, JavaScript, HTML, CSS
- **Databases & Data:** MySQL, PostgreSQL, SQLite, data modeling, ACID principles
- **Systems & Infrastructure:** Linux/Unix, Azure, Docker, networking fundamentals, cloud deployment
- **Tools & Practices:** Git/GitHub, REST APIs, Agile/Scrum, VS Code, R-Studio

WORK EXPERIENCE

Online ESL Instructor | Native Camp & VIPKid, (Japan & China) *2019 – Present*

- Delivered 11,000+ remote lessons
- Developed strong communication and client relationship skills

University Administrative Assistant | Cambridge/UTPL, Ecuador *2016 – 2020*

- Managed student information systems and academic records
- Provided technical support for online learning platforms
- Maintained data integrity across academic systems

ESL Teacher | English House, Ecuador *2016 – 2020*

- Taught English across CEFR levels A1–C2 including TOEFL and IELTS preparation
- Collaborated with institutional programs for structured language training